Comments on "Self-Targeting in U.S. Transfer Programs"

This is an ambitious paper on an interesting question, which is whether take-up in U.S. transfer programs is positively or negatively selected on need--whether the neediest households are the most likely to take-up benefits or the opposite. Despite what seems to be a large number of researchers who think it is negatively selected on need, this reviewer agrees with the statement in the introduction to this paper--the evidence is mixed, at best, that that is the case.

The basic idea in this paper is to measure "need" by consumption, not income or anything else. And to show that you get positive selection, when you measure need by consumption.

This referee report will discuss a number of important questions about the analysis--alternative interpretations of the results, the lack of attention to the large body of literature on consumption smoothing and the insurance value of transfers, and important issues concerning measurement error in income and in eligibility for transfer programs.

But a first question: why are the authors using the PSID and not the Consumer Expenditure Survey (CE)? The CE is universally used in the US for the calculation of consumption. It asks very detailed questions on consumption, far more than the PSID, is available for many years, and does not have the sample size problems that the PSID does. The PSID is subject to unknown attrition bias, having suffered huge attrition in its first generation--those alive when it started in 1968--with consequently unknown attrition bias in the second and third generations that are now part of the sample. Yes, the paper wants to do lifetime income, and panel data are needed for that, but that could be a supplementary analysis. The PSID has slightly better data on transfer program receipt than the CE, but it's unclear if that outweighs the CE's advantages.

Interpretation of Main Results

The story of the paper is really all in Table 2, Panel A: receipt of transfers holding Income (Y: this is pretax pretransfer income) fixed is positively selected on consumption--those with higher rates of receipt have higher consumption.

One odd feature of the analysis is the sole focus on selection on consumption rather than on Y. The government uses Y, and it positively selects on Y: receipt is lower, the higher the income. My impression is that this is the main measure of selection that is used in the literature: whether those with greater need, measured by pretax pretransfer income, are more likely to participate. Table 2 shows that this is already the case in the paper's data, so the paper is already finding positive selection on need. Yes, it is even more positively selected when looking at consumption conditional on income, so the paper is just finding that selection is more positive than we thought.

But an immediate problem is that income is measured with a high degree of error--and greater error than consumption, as the authors correctly note the literature shows. So the lower selection on income than on consumption shown in Panel A could be entirely the result of measurement error in income. More on this below.

Consumption vs Income and Consumption Smoothing

Although one hypothesis for the difference in selection on income and consumption in Panel A is measurement error in income, it may be that the paper's interpretation is right: that low income households do consumption smoothing (that is not directly stated in the paper but it seems to be the interpretation, especially in the use of lifetime income as a ranking variable). It is rather surprising that a paper focused on the differences in income and consumption in the low income population has no discussion of the enormous literature on consumption smoothing, liquidity constraints, precautionary saving, insurance value of transfers, etc.

There are many researchers who work on the low income population who think that low income households are completely liquidity constrained--they cannot borrow, they do not save, etc. If that were the case, income in a period (properly measured) would equal consumption in the period. The data here reject that, but it could be the result of measurement error in income. But the paper rejects that hypothesis in its analysis of measurement error in income--it takes the difference as true (see below for that analysis).

But take, first, the literature on precautionary saving (Hubbard-Skinner-Zeldes JPE 1995 is the seminal reference). Transfer programs reduce that saving. Further, there is heterogeneity across households in the degree to which they do that. Those who do not save will have less liquid assets in the future and will rely more on transfer programs. So those in the bottom quintile of consumption in the receipt of programs may be those who didn't save.

Likewise, on the insurance side, low income families have alternative sources of insurance. There is a large, unreferenced literature on the insurance value of transfers (examples are Blundell et al., AER, 2008; Blundell-Pistaferri, JHR, 2003). Implicit in that literature is that transfers may not have full insurance value.

In both of these cases, the problem is that one needs a counterfactual: how would consumption today (and last year?) be different in the absence of transfers? One needs a behavioral model for that.

And the larger problem is that consumption is not a very good measure of need--which this paper assumes it to be. It is endogenous and chosen by households with life cycle considerations and in light of what liquidity constraints they face. It it not a measure of resources; it is a measure of how households allocate their resources across different periods, according to their tastes and preferences. Most analysts of poverty think that a measure of exogenous resources should be used to measure need, not an endogenous variable like consumption.

Measurement Error in Y

There are two important measurement error issues in the use of PSID pretax, pretransfer annual reported income as a measure of need.

One is that the annual income measure in the PSID--even if reported accurately--is not what agencies use to establish eligibility and benefits. In the simplest cases, they use monthly income, which is different than PSID annual income, and it is very likely that households participate in programs only some months in a year. But it's even more complicated than that, because most transfer programs use some combination of past monthly income, an estimate of future income, etc. to calculate eligibility. It is almost impossible to capture this with the PSID.

Now, it may in truth be the case that program operators collect low quality income data when they assess eligibility. True monthly income may equal monthly consumption exactly, for example. In that case, the policy implication is that transfer programs should do a better job of collecting income data (e.g., using register data, etc.).

The other obvious problem is that PSID repondents may underreport Y income to the PSID interviewers, either intenttionally or unintentionally. There are many papers which match administrative earnings data to survey data on earnings and find large differences (e.g., Bollinger et al., JPE, 2019).

This paper addresses that problem by using predicted Y in place of actual Y, using a prediction equation from a different data set (the CPS). So it uses IV, which is a textbook approach to correcting for measurement error. However, IV requires that all the variables used to predict are not direct determinants of the outcome in question (consumption) and that they are exogenous. The variables used as instruments here--hours of work, weeks worked, self-employment status, etc.)--are unlikely to satisfy either condition. A much more sophisticated treatment of measurement error in income is needed.

Miscellaneous Comments

Table 1 take-up rates are lower than in official and other estimates. The paper notes this and has some of the alternate estimates in Appendix Table A10. But no explanation is given for the difference.

Housing is a special case: it is not an entitlement program because only a certain number of housing units and vouchers are funded by the government. It is rationed, and there are long waiting lists for a unit or voucher. Yes, self-selection can occur by who applies and who does not, but the government has its own priority list for who to select from the queue. If it is the government that is doing positive selection, perhaps the policy recommendation is that all programs should be rationed and we should let the government decide who is allowed to participate.